

Doubly robust machine learning

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Warm-up

Standard semiparametrics

Debiased GMM

Partially linear model

- Consider partly linear model $Y = D\theta_0 + \gamma_0(X) + \epsilon$, write as moment condition $E[D(Y - D\theta - \gamma(X))] = 0$, where $\gamma_0(X) = [Y - D\theta_0 \mid X]$.
- Consider a plug-in estimator based on preliminary estimate $\hat{\gamma}$,

$$\hat{\theta}_{PI} = \frac{E_n[D(Y - \hat{\gamma}(X))]}{E_n[D^2]}$$

Then

$$\hat{\theta}_{PI} - \theta_0 = \frac{E_n[D(Y - \theta_0 D - \hat{\gamma}(X))]}{E_n[D^2]} = \frac{E_n[D\epsilon]}{E_n[D^2]} + \frac{E_n[D(\gamma_0(X) - \hat{\gamma}(X))]}{E_n[D^2]}$$

- First term standard, second term potentially non-zero, there is bias

$$B = E[D(\gamma_0(X) - \hat{\gamma}(X))],$$

which we need to be of order $o_p(1/\sqrt{n})$. Usually need to analyze case-by case.

- Suppose we use a series estimator (that is, ordinary least squares (OLS)): $\hat{\gamma}(X) = X'\hat{\delta}$, and $\gamma_0(X) = X'\delta_0 + r_\gamma(X)$. Then

$$\begin{aligned} B &= E[D(\gamma_0(X) - \hat{\gamma}(X))] = E[Dr_\gamma(X)] + E[X'(\delta_0 - \hat{\delta})] \\ &= E[r_\alpha(X)r_\gamma(X)] + 0 \end{aligned}$$

where $E[D | X] = \alpha(X) = X'\rho + r_\alpha(X)$ is propensity score.

- So series is **doubly robust** (as we discussed in OLS lecture)

- What if we use something fancier, say the lasso? Suppose we sample split. Then

$$B = E[D(\gamma_0(X) - \hat{\gamma}(X))] = E[\pi(X)E[(\gamma_0(X) - \hat{\gamma}(X)) \mid X]].$$

We run into trouble if bias correlated with propensity score.

- Suppose X scalar, and no approximation error. Then optimal to set $\hat{\delta} = 0$ from MSE perspective if $\delta^2 < \sigma_\epsilon^2/nE[X_i^2]$. But if we do this, we have bias equal to $B = E[X^2]\pi\delta \asymp 1/\sqrt{n}$.

Double machine learning to the rescue

- Use instead a **doubly robust moment condition**

$$E[(D - \alpha(X))(Y - D\theta - \gamma(X))],$$

which we can think of as debiasing the original sample moment condition

- Leads to estimator

$$\hat{\theta}_{DB} = \frac{E_n[(D - \hat{\alpha}(X))(Y - \hat{\gamma}(X))]}{E_n[D(D - \hat{\alpha}(X))]},$$

and bias

$$B \asymp \sqrt{n}E[(\alpha_0(X) - \hat{\alpha}(X))(\gamma_0(X) - \hat{\gamma}(X))] \preceq \sqrt{n}\|\hat{\alpha} - \alpha_0\| \|\hat{\gamma} - \gamma_0\|.$$

OK if our estimators of α_0 and γ_0 converge faster than $n^{-1/4}$.

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- A plethora of causal/structural parameters in economics can be expressed via **semiparametric** moment conditions $E[g(Z, \gamma_0, \theta_0)] = 0$, where γ_0 is a nuisance function (conditional mean, quantile, density etc), and θ_0 parameter of interest.
- Make two simplifications:
 1. Data is $Z = (Y, X)$, and $\gamma_0(X) = E[Y | X]$
 2. $g(Z, \gamma, \theta) = m(Z, \gamma) - \theta$, with $m(Z, \gamma) - m(Z, 0)$ linear in γ .

m linear in γ allows for estimation based on **doubly robust** moments, rather than just **locally robust/orthogonal** moments (Chernozhukov et al. [2022](#)), and simpler regularity conditions, but most theory goes through without these simplifications

$\theta_0 = E[m(Z, \gamma_0)]$ where $\gamma_0(X) = E[Y | X]$

- Running example: $X = (D, W)$, and $\theta_0 = E[\gamma_0(1, W)]$
 - Under unconfoundedness (i.e. $E[Y(1) | D = 1, W] = E[Y(1) | W]$), $\theta_0 = E[Y(1)]$.
 - Inference on the ATE just slightly more complicated, here $\theta_0 = E[\gamma_0(1, W) - \gamma_0(0, W)]$.
- Other examples: Average effect of shifting distribution of covariates
 $\theta_0 = \int \gamma_0(x) dF_1(x) - E[\gamma_0(X)]$ (Stock [1989](#)), average derivative $\theta_0 = E[\partial \gamma_0(D, W) / \partial D]$
...hundreds of papers applying general theory to different settings

Plug-in approach

- Simplest approach: estimate γ_0 nonparametrically, and then set

$$\hat{\theta}_{PI} = \frac{1}{n} \sum_{i=1}^n m(Z_i, \hat{\gamma}) = \frac{1}{n} \sum_{i=1}^n \hat{\gamma}(1, W_i).$$

- Then

$$\sqrt{n}(\hat{\theta}_{PI} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\hat{\gamma}(1, W_i) - \gamma_0(1, W_i)) + \frac{1}{\sqrt{n}} \sum_{i=1}^n (\gamma_0(1, W_i) - \theta_0).$$

Second term well-behaved and asymptotically normal (corresponds to an oracle), but first typically non-zero mean, since $\hat{\gamma}$ biased for γ

- Need bias to be $o_p(n^{-1/2})$, can be hard to ensure (typically check on case-by-case basis), and doesn't hold if e.g. $\hat{\gamma}$ is estimated via the lasso $\implies \hat{\theta}_{PI}$ asymptotically biased.

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Debiasing: Key insight!

- Notice that for any γ ,

$$E[\gamma(1, W)] = E\left[\frac{D}{\pi(W)}\gamma(X)\right] \implies E[\gamma(1, W) - \gamma_0(1, W)] = E\left[\frac{D}{\pi(W)}(\gamma(X) - Y)\right].$$

where $\pi(W) = E[D \mid W]$ is propensity score.

- Suggests estimating bias as $\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\pi(W_i)} (\hat{\gamma}(X_i) - Y_i)$
- If $\hat{\gamma}$ estimated in separate sample and $\pi(W_i)$ known, then subtracting off $\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\pi(W_i)} (\hat{\gamma}(X_i) - Y_i)$ would yield estimator of θ_0 that is unbiased conditional on $\hat{\gamma}$.

- Debiased generalized method of moments (GMM) subtracts off feasible bias estimate,

$$\hat{\theta}_{DB} = \frac{1}{n} \sum_{i=1}^n \hat{y}_{\ell(i)}(1, W_i) - \frac{1}{n} \sum_{i=1}^n \hat{\alpha}_{\ell(i)}(X_i) (\hat{y}_{\ell(i)}(1, W_i) - Y_i)$$

where $\hat{y}_{\ell(i)}$ estimated on sample **excluding** i and $\hat{\alpha}$ is some estimator of $\alpha(X) := D/\pi(W)$.

- To avoid estimating γ_0 n times, usually split sample into $L = 5$ or $L = 10$ folds, and $\ell(i)$ excludes fold that i belongs to.
- This **cross-fitting** avoids own observation bias analogous to the jackknife instrumental variables estimator (JIVE) in instrumental variables (IV) settings.

- Since $\theta(\gamma) = E[m(Z, \gamma)]$ is a linear functional, if the functional is continuous (i.e. $E[m(W, \gamma)] \leq C\|\gamma\| =: CE[\gamma(W)^2]$ for all $\gamma \in L^2$), by the Riesz representation theorem, there exists $\alpha_0 \in L^2$ such that for all $\gamma \in L^2$

$$E[m(Z, \gamma)] = E[\alpha_0(X)\gamma(X)] \quad (1)$$

Called **Riesz representer (RR)**. Existence of RR equivalent to semiparametric variance bound being finite (Hirshberg and Wager 2021; Newey 1994), i.e. $\sqrt{n}$ -consistent estimation possible

- In our case, $\alpha(X) = D/\pi(W)$. General recipes for RR construction available (e.g. Chernozhukov et al. 2022; Chernozhukov et al. 2018)

- Debiased GMM equivalent to GMM based on moment condition

$$\theta_0 = E[\psi(Z, \gamma_0, \alpha_0)], \quad \psi(Z, \gamma, \alpha) = m(Z, \gamma) + \alpha(X)(Y - \gamma(X))$$

In our case $\psi(Z, \gamma, \alpha) = \gamma(1, W) + (Y - \gamma(X))D/\pi(W)$.

- Moment condition **doubly robust**

$$\begin{aligned} E[\psi(Z, \gamma, \alpha)] - \theta_0 &= E[m(Z, \gamma) - m(Z, \gamma_0) + \alpha(X)(Y - \gamma(X))] \\ &=_{(1)} E[m(Z, \gamma) - m(Z, \gamma_0) + \alpha(X)(\gamma_0(X) - \gamma(X))] \\ &=_{(2)} E[(\alpha(X) - \alpha_0(X))(\gamma_0(X) - \gamma(X))] \end{aligned}$$

where (1) uses definition of γ_0 and (2) uses definition of RR, $E[m(Z, \gamma)] = E[\alpha_0(X)\gamma(X)]$.

- Idea of using doubly robust moments old (Robins, Rotnitzky, and Zhao [1994](#), [1995](#)).
Innovation of debiased GMM is to combine it with cross-fitting, which allows for weaker regularity conditions on estimator of γ that accommodate regularized estimators (lasso, random forests, neural nets etc).
- Price: need to estimate α as well as γ , and it could be that α is non-smooth and hard to estimate
- Existence of RR allows for multiple approaches to estimating θ_0 : regression (plug-in), propensity score weighting, and doubly-robust moments:

$$\theta_0 = E[m(Z, \gamma_0)] = E[\alpha_0(X)Y] = E[\psi(Z, \gamma_0, \alpha_0)].$$

- Simple conditions for asymptotic normality of

$$\hat{\theta}_{DB} = \frac{1}{n} \sum_{i=1}^n [m(Z_i, \hat{\gamma}_{\ell(i)}) + \hat{\alpha}_{\ell(i)}(X_i)(Y_i - \hat{\gamma}_{\ell(i)}(W_i))], \text{ since}$$

$$\sqrt{n}(\hat{\theta}_{DB} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (R_{1i} + R_{2i} + R_{3i}) + \frac{1}{\sqrt{n}} (\psi(Z_i) - \theta_0),$$

with

$$R_{1i} = m(Z_i, \hat{\gamma}_{\ell(i)}) - m(Z_i, \gamma_0) - \alpha_0(X_i)(\hat{\gamma}_{\ell(i)}(X_i) - \gamma(X_i))$$

$$R_{2i} = (\hat{\alpha}_{\ell(i)}(X_i) - \alpha_0(X_i))(Y_i - \gamma_0(X_i))$$

$$R_{3i} = -(\hat{\alpha}_{\ell(i)}(X_i) - \alpha_0(X_i))(\hat{\gamma}_{\ell(i)}(X_i) - \gamma(X_i))$$

Lemma 8 in Chernozhukov et al. 2022

$\sqrt{n}(\hat{\theta}_{DB} - \theta_0) \xrightarrow{d} \mathcal{N}(0, E[\psi(Z_i)^2])$, provided

1. First-step estimators consistent: $\|\hat{\alpha} - \alpha\| + \|\hat{\gamma} - \gamma\| = o_p(1)$
2. $\max_i \text{var}(Y_i | X_i) < \infty$ and $\max_i \alpha(X_i) < \infty$ (strong overlap)
3. $\|\hat{\alpha} - \alpha_0\| \|\hat{\gamma} - \gamma_0\| = o_p(n^{-1/2})$:

- Conditions just ensure all three terms R_{1i}, R_{2i}, R_{3i} asymptotically negligible.
- Condition 3: **Only product of biases needs to be small!** In contrast, with plug-in approach needed bias of $\hat{\gamma}$ small. Allows for model selection / machine learning.

The following result is a variant on Theorem 4 in Newey and Robins (2018), and on Theorem 8 and Corollary 9 in Ichimura and Newey (2017):

Lemma

Suppose that $E[m_\Delta(Z_i, \gamma)^2] \leq K\|\gamma\|^2$ and $\text{var}(Y_i | X_i) \leq K$. If the series approximation is a splines, and we cross-fit

$$\sqrt{n}(\hat{\theta}_{PI} - \theta_0) - \frac{1}{\sqrt{n}} \sum_i \psi(Z_i, \gamma_0, \alpha_0, \theta_0) \preceq \sqrt{p \log(p)/n} (1 + \sqrt{p}\|r_\gamma\|) + \Delta_n^*.$$

where $\Delta_n^* = \sqrt{p/n} + \|r_\alpha\| + \|r_\gamma\| + \sqrt{n}\|r_\delta\|\|r_\gamma\|$.

Let $\hat{y}_\ell(x_i) = x_i' \hat{\delta}_\ell$. Let $\tilde{\delta}_\ell = \hat{\delta}_\ell - \delta$, and let $\epsilon_i = Y_i - \gamma_0(X_i)$. Normalize $E[XX'] = I$. Then we can decompose

$$\sqrt{n}(\hat{\theta}_{PI} - \theta_0) - \frac{1}{\sqrt{n}} \sum_i \psi(Z_i, \gamma_0, \alpha_0, \theta_0) = \frac{1}{\sqrt{n}} \sum_\ell R_{1\ell} + \frac{1}{\sqrt{n}} \sum_\ell R_{2\ell}$$

where

$$R_{1\ell} = \sum_{i \in \ell} (m(Z_i, \hat{y}_\ell) - m(Z_i, \gamma_0) - E_{F_X}[\alpha_0 \tilde{y}_\ell]) = \sum_{i \in \ell} (m_\Delta(Z_i, \tilde{y}_\ell) - E_{F_X}[\alpha_0 \tilde{y}_\ell])$$

$$R_{2\ell} = \sum_{i \in \ell} (E_{F_X}[\alpha_0 \tilde{y}_\ell] - \alpha_0(X_i) \epsilon_i)$$

The term $R_{1\ell}$ is mean zero conditional on $\mathcal{N} \setminus \ell$, with variance bounded by $\sum_{i \in \ell} E[m_\Delta(Z_i, \tilde{y}_\ell)^2 \mid \mathcal{N} \setminus \ell] \leq K \|\tilde{y}_\ell\|^2 n/L$, so that $n^{-1/2} \sum_\ell R_{1\ell} \preceq L^{-1/2} \sum_\ell \|\tilde{y}_\ell\|^2$ (without cross-fitting, we can bound R_{1i} using the results from Andrews [1994](#)).

Decompose the second term further,

$$\frac{1}{\sqrt{n}} \sum_{\ell} R_{2\ell} = \frac{\sqrt{n}}{L} \sum_{\ell} \rho'_0 \tilde{\delta}_{\ell} - \frac{1}{\sqrt{n}} \sum_i \alpha_0(X_i) \epsilon_i - \sqrt{n} E[\alpha_0(X) r_Y(X)],$$

where we use $\rho_0 = E[\alpha_0 X]$. By Lemma 4.1 in Belloni et al. (2015),

$$\sqrt{N} \rho'_0 \tilde{\delta}_{\ell} = N^{-1/2} \sum_{i \notin \ell} \rho'_0 X_i \epsilon_i + O_p(\sqrt{p \log(p)/N} (1 + \sqrt{p} \ell_p \|r_Y\|) + \ell_p \|r_Y\|) \text{ with } \ell_k = 1 + \log(k)$$

for polynomials ℓ_k constant for splines, where $N = n(1 - 1/L)$. Therefore, with a fixed number of folds,

$$\frac{\sqrt{n}}{L} \sum_{\ell} \rho'_0 \tilde{\delta}_{\ell} = \frac{1}{\sqrt{n}(L-1)} \sum_{\ell} \sum_{i \notin \ell} \rho'_0 X_i \epsilon_i + O_p(\sqrt{p \log(p)/n} (1 + \sqrt{p} \ell_p \|r_Y\|) + \ell_p \|r_Y\|)$$

Since $\sum_{\ell} \sum_{i \notin \ell} \rho'_0 X_i \epsilon_i = (L-1) \sum_i \rho'_0 X_i \epsilon_i$, and since $E[\alpha_0(X_i) r_Y(X_i)] = E[r_{\delta} r_Y]$, because r_Y is orthogonal to linear functions of X , it follows that for splines,

$$\frac{1}{\sqrt{n}} \sum_{\ell} R_{2\ell} \asymp \frac{1}{\sqrt{n}} \sum_i r_{\alpha}(X_i) \epsilon_i + \sqrt{p \log(p)/N} (1 + \sqrt{p} \|r_Y\|) + \|r_Y\| + \sqrt{n} \|r_{\delta}\| \|r_Y\|.$$

Since $\|\tilde{y}_{\ell}\| \asymp \|r_Y\| + \sqrt{p/n}$ by standard arguments, the result follows.

- Debiasing plug-in estimator can be done in many ways, see e.g., Laan and Rubin (2006), Laan and Rose (2018), Hirshberg and Wager (2021), and Athey, Imbens, and Wager (2018). I'm not clear on relative merits of different approaches.
- Cross-fitting not needed for estimating β in partially linear model $Y = D\beta + g(W) + U$ (Donald and Newey 1994)
 - This is why post double lasso (Belloni, Chernozhukov, and Hansen 2014) doesn't need to cross-fit
 - Reason for this is that there is no endogeneity, $E[Y - \gamma_0(X) \mid Z] = 0$, so no own observation bias from estimating γ_0 . In general, cross-fitting not needed (at least with series!) if there is not endogeneity in this sense.
 - Otherwise, cross-fitting solves own observation bias. Generalization of jackknife IV to other settings.

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